CONSOLE TIC TAC TOE USING C++

Submitted In Partial Fulfilment of Requirement of The Degree Of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

RKDF UNIVERSITY, RANCHI

BY

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B.Tech. CSE-IV

Jan-June 2026

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to RKDF University, Ranchi, Jharkhand for providing us the opportunity to carry out this Mini project as a part of the curriculum of the Bachelor of Engineering in Computer Science Engineering. We convey our deep and sincere thanks to our Head of Department, Mrs. Kirti Verma, for her support, encouragement, valuable suggestions, and constant motivation throughout the duration of this project. We are extremely thankful and indebted to our project guide, Mr. Abhisekh Kumar Singh, for his continuous guidance, constant support, and for providing his expertise that greatly assisted the successful completion of this work. We would also like to express our heartfelt thanks to all faculty members of the Department of CSE for their help and useful feedback. Finally, we extend our thanks to our friends and family for their encouragement, moral support, and understanding throughout this project development.

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1. INTRODUCTION

Introduction to the Internship

The internship plays an important role in enhancing the practical knowledge and technical skills of students. It provides an opportunity to understand real-world working environments and apply theoretical concepts learned during academic studies. As a part of the B.Tech Computer Science & Engineering curriculum, I completed my internship at WingsIT Education and Solutions, a reputed computer training and skill development institute located in Jharkhand.

The internship was conducted in the field of C++ Development from 04 April 2026 to 20 May 2026. The training program was designed to improve programming skills, logical thinking, and software development knowledge through practical implementation and project-based learning.

During the internship period, I learned various concepts of C++ programming such as variables, loops, functions, arrays, conditional statements, object-oriented programming concepts, and problem-solving techniques. The internship also provided exposure to real-time coding practices, debugging methods, and project implementation strategies.

As a part of the internship project, I developed a Console-Based Tic Tac Toe Game using C++. The project is a simple two-player game that runs in the command-line interface and demonstrates the use of programming fundamentals including loops, functions, arrays, conditions, and user-defined logic. The project helped in strengthening my understanding of game logic development and console application programming.

The internship experience at WingsIT Education and Solutions was highly beneficial and helped me improve my technical abilities, communication skills, teamwork, and confidence in software development. It also provided industry-oriented exposure that will be useful for my future academic and professional career.

About the Organization – WingsIT Education and Solutions

WingsIT Computer Training Center is a highly reputed IT education and skill development institute established in the year 2012. The institute is government registered under the Companies Act and is also ISO certified, ensuring quality education and reliable training standards.

The institute specializes in providing technical, professional, and career-oriented computer education to students, job seekers, and professionals. WingsIT has successfully trained more than 10,000 students and is recognized as an authorized training partner of Tally and NIELIT.

The organization offers a wide range of courses including:

- **ADCA (Advanced Diploma in Computer Applications)**
- **DCA (Diploma in Computer Applications)**
- **Tally with GST**
- **Basic Computer Courses**
- **CCC (NIELIT)**
- **Graphic Designing**
- **Accounting & Office Automation**
- **Programming and IT Skills**

The institute is equipped with modern infrastructure including smart classrooms, computer laboratories, air-conditioned learning environments, and practical training facilities. The faculty members are experienced, supportive, and focused on providing concept clarity through hands-on learning methods.

WingsIT also conducts various student development programs, workshops, scholarship activities, and skill enhancement sessions to prepare students for real-world industry requirements. The organization is widely appreciated for its affordable fee structure, quality technical education, and career-oriented training approach.

2. OBJECTIVE OF INTERNSHIP

The main objective of the internship was to gain practical knowledge and real-world experience in the field of programming and software development. The internship helped in understanding how theoretical concepts learned during academic studies are applied in actual project development environments.

The internship mainly focused on improving programming skills in C++ and enhancing logical thinking and problem-solving abilities through practical assignments and project implementation. During the training period, special emphasis was given to learning the concepts of Object-Oriented Programming (OOP) and their implementation in software development.

The objectives of the internship were as follows:

- **To develop a strong understanding of C++ programming language.**
- **To learn and apply Object-Oriented Programming (OOP) concepts.**
- **To improve logical thinking and problem-solving skills.**
- **To gain practical exposure to software development techniques.**
- **To understand program structure, debugging, and error handling.**
- **To develop a console-based application using C++.**
- **To enhance confidence in coding and project implementation.**
- **To improve self-learning and independent project development abilities.**
- **To gain knowledge about real-time programming practices and project workflow.**

As a part of the internship, an individual project titled "Console Tic Tac Toe Using C++" was developed. The project aimed to create a simple two-player game using fundamental programming concepts such as loops, arrays, functions, conditional statements, and game logic implementation. Through this project, practical knowledge of program design, implementation, testing, and debugging was achieved successfully.

3. COMPANY OVERVIEW

Overview of WingsIT Education and Solutions

WingsIT Education and Solutions is a reputed computer training and skill development institute established in the year 2012 with the objective of providing quality technical education and professional training to students and job seekers. The organization is known for its practical learning approach, modern training methods, and industry-oriented educational environment.

The institute is a government-registered organization under the Companies Act and is also ISO certified, which reflects its commitment toward maintaining educational standards and providing reliable training services. Over the years, WingsIT has successfully trained thousands of students and has built a strong reputation in the field of computer education and technical skill development.

The organization focuses on developing technical knowledge, professional confidence, and practical skills among students through hands-on training and project-based learning. WingsIT Education and Solutions believes in bridging the gap between academic learning and industry requirements by providing real-world exposure and practical implementation experience.

Infrastructure and Learning Environment

WingsIT Education and Solutions provides a modern and student-friendly learning environment. The institute is equipped with well-maintained computer laboratories, smart classrooms, and modern teaching facilities that support practical learning and technical training.

The classrooms are designed to provide a comfortable and interactive atmosphere for students. Practical sessions are conducted regularly to ensure that students gain proper hands-on experience in programming, software development, and technical implementation.

The institute also maintains a disciplined and professional environment that motivates students to improve their technical and communication skills. The availability of experienced faculty members and supportive mentors helps students understand concepts more effectively and encourages them to solve practical problems independently.

Training Methodology

The training methodology followed by WingsIT Education and Solutions mainly focuses on practical knowledge and skill development. Instead of limiting learning to theoretical concepts only, the organization encourages students to work on live projects, coding assignments, practical implementations, and problem-solving activities.

The institute emphasizes:

- **Practical learning approach**
- **Hands-on programming sessions**
- **Real-time project implementation**
- **Logical and analytical thinking**
- **Problem-solving techniques**
- **Industry-oriented technical training**
- **Student interaction and mentorship**

This methodology helps students gain confidence in their technical abilities and prepares them for future academic and professional challenges.

Faculty and Student Support

One of the major strengths of WingsIT Education and Solutions is its experienced and supportive faculty team. The trainers focus on concept clarity, individual attention, and practical implementation of technical topics. Students are guided throughout the training process and are encouraged to improve their coding skills through continuous practice and project work.

The institute also supports students through doubt-clearing sessions, mentorship programs, workshops, and technical guidance activities. This supportive environment helps students improve their overall learning experience and professional development.

Reputation and Achievements

WingsIT Education and Solutions has earned a positive reputation for providing affordable and quality technical education. The organization is widely appreciated for its transparent training system, practical-oriented teaching methods, and student satisfaction.

The institute has contributed significantly toward technical skill development and career preparation of students in the field of information technology and computer applications. Its focus on practical knowledge, discipline, and professional growth has made it a trusted name in computer education and training.

4. INTERNSHIP ROLE & RESPONSIBILITY

Role During Internship

During the internship at WingsIT Education and Solutions, I worked as a trainee in the field of C++ Development. My primary role was to learn programming concepts, improve coding skills, and implement practical applications using the C++ programming language. The internship mainly focused on enhancing logical thinking, understanding Object-Oriented Programming concepts, and developing problem-solving abilities through practical implementation.

I actively participated in coding sessions and project development activities under the guidance of my mentor. The internship provided me with an opportunity to understand the real-time workflow of software development and practical programming techniques.

Responsibilities Performed

During the internship period, I performed several technical and learning-related responsibilities. These responsibilities helped me gain practical exposure and improve my programming knowledge.

My major responsibilities during the internship were:

- **Learning and understanding the fundamentals of C++ programming.**
- **Studying and implementing Object-Oriented Programming (OOP) concepts.**
- **Writing and testing C++ programs for logical problem-solving.**
- **Developing the Console Tic Tac Toe Game project individually.**
- **Identifying and correcting programming errors through debugging.**
- **Improving coding efficiency and logical thinking skills.**
- **Practicing programming concepts through regular coding exercises and practical sessions.**

Mentor Guidance and Support

Throughout the internship, guidance and support were provided by the mentor and faculty members of WingsIT Education and Solutions. The mentor helped in understanding programming concepts, project implementation techniques, and debugging methods. Regular feedback and technical suggestions were provided to improve coding practices and problem-solving approaches.

The mentorship and supportive learning environment played an important role in the successful completion of the internship project and overall technical skill development.

Individual Contribution

The project titled “Console Tic Tac Toe Using C++” was developed individually as part of the internship training. All coding, logic implementation, testing, and debugging tasks were completed independently under mentor supervision. This helped in improving self-learning capability, programming confidence, and independent project development skills.

5. INTERNSHIP ACTIVITIES AND PROJECTS

Internship Activities

During the internship at WingsIT Education and Solutions, various technical and practical learning activities were performed to improve programming knowledge and software development skills. The internship mainly focused on enhancing coding abilities, logical thinking, and understanding of Object-Oriented Programming concepts using the C++ programming language.

Regular practical sessions were conducted to strengthen programming fundamentals and improve problem-solving approaches. Different coding exercises and program implementations were practiced throughout the internship period. The activities helped in understanding the structure of programs, debugging techniques, and the execution flow of console-based applications.

The major activities performed during the internship included:

- **Learning C++ programming fundamentals.**
- **Understanding Object-Oriented Programming (OOP) concepts.**
- **Writing and executing C++ programs.**
- **Performing logical problem-solving exercises.**
- **Practicing conditional statements, loops, arrays, and functions.**
- **Implementing console-based project logic.**
- **Testing and improving program performance.**
- **Correcting syntax and logical errors during execution.**

Project Undertaken

As a part of the internship training, a project titled "Console Tic Tac Toe Using C++" was developed individually. The project is a console-based two-player game developed using the C++ programming language. The game was designed to demonstrate the practical implementation of programming concepts learned during the internship.

The Tic Tac Toe game allows two players to play alternately by entering their choices on the console screen. The program checks game conditions after every move and determines the winner or draw result automatically.

The project was developed using fundamental programming concepts such as:

- **Arrays**
- **Functions**
- **Conditional Statements**
- **Loops**
- **User Input Handling**
- **Game Logic Implementation**
- **Object-Oriented Programming Concepts**

Project Development Process

The development of the project was completed in multiple stages. Initially, the game structure and logic flow were planned. After that, coding was started by implementing the game board, player input system, move validation, and winner detection logic.

Testing and debugging were performed regularly during development to ensure smooth execution of the program. Different input combinations were tested to verify the correctness of the game logic and output results.

The project development process included:

1. **Planning the game structure and rules.**
2. **Designing the console game interface.**
3. **Implementing player input functionality.**
4. **Creating game board display logic.**
5. **Implementing winner and draw checking conditions.**
6. **Testing different gameplay scenarios.**
7. **Debugging and improving program execution.**

Outcome of the Project

The successful completion of the Console Tic Tac Toe project helped in improving programming confidence and practical implementation skills. The project provided hands-on experience in software development and strengthened understanding of C++ programming concepts.

The internship project also improved:

- **Logical thinking ability**
- **Problem-solving skills**
- **Debugging techniques**
- **Program design understanding**
- **Independent coding capability**
- **Confidence in project development**

The project served as an effective learning experience and provided practical exposure to console application development using C++.

6. SKILLS AND LEARNING OUTCOME

Skills Developed During Internship

The internship at WingsIT Education and Solutions helped in developing both technical and personal skills. Through continuous practice, project implementation, and mentor guidance, significant improvement was achieved in programming knowledge and practical problem-solving abilities.

One of the major skills improved during the internship was logical thinking and logic building. Working on programming problems and implementing the Tic Tac Toe project helped in understanding how to design program logic and solve technical challenges effectively.

The internship also enhanced coding efficiency and understanding of C++ programming concepts through practical implementation and debugging activities.

The major skills developed during the internship are listed below:

- **Logical thinking and logic building**
- **C++ programming skills**
- **Object-Oriented Programming (OOP) concepts**
- **Problem-solving ability**
- **Debugging and error handling**
- **Program design and implementation**
- **Understanding of console-based application development**
- **Independent project development skills**
- **Analytical thinking capability**

Learning Outcomes

The internship provided valuable practical exposure and helped in understanding how programming concepts are applied in real-world project development. Through continuous coding practice and project implementation, theoretical concepts became easier to understand and apply practically.

The development of the Console Tic Tac Toe project improved understanding of loops, functions, arrays, conditions, and game logic implementation. It also helped in learning how to manage program flow and improve code structure for better execution.

During the internship, knowledge was gained about:

- **Real-time programming practices**
- **Practical implementation of C++ concepts**
- **Project planning and execution**
- **Software testing and debugging methods**
- **Logical problem-solving approaches**
- **Console application development techniques**
- **Importance of structured coding practices**

Overall Experience and Improvement

The internship experience was highly beneficial for both academic and technical growth. It helped in improving confidence in programming and increased interest in software development and problem-solving activities.

Working independently on the project also enhanced self-learning capability and provided practical experience in handling programming challenges. The internship created a strong foundation for future learning in software development and advanced programming concepts.

7. PROJECT TITLE AND OBJECTIVE

Project Title

Console Tic Tac Toe Using C++

Project Objective

The main objective of this project was to develop a simple and interactive console-based Tic Tac Toe game using the C++ programming language. The project was designed to strengthen programming knowledge and provide practical implementation experience of fundamental programming concepts and Object-Oriented Programming principles.

The project aimed to create a two-player game where users can play turn by turn on a 3×3 game board through the console interface. Another objective of the project was to improve logical thinking, problem-solving ability, and understanding of game logic implementation.

The specific objectives of the project were:

- **To implement a console-based gaming application using C++.**
- **To understand the practical use of arrays, loops, and functions.**
- **To apply conditional statements and game decision logic.**
- **To improve debugging and error-handling skills.**
- **To enhance logical thinking and programming confidence.**
- **To develop a user-friendly game interface in console mode.**
- **To understand turn-based game execution and winner detection mechanisms.**

Project Description

The Console Tic Tac Toe Using C++ project is a two-player console game developed using standard C++ programming concepts. The game allows two players to play alternately using keyboard inputs on a 3×3 game board.

The project uses arrays to store player moves and functions to manage different game operations such as drawing the board, taking player input, checking game results, and controlling gameplay execution.The game automatically checks for winning conditions after every move and displays the result when either player wins or when the game ends in a draw.

The project includes the following major functionalities:

- **3×3 game board creation**
- **Player turn management**
- **Keyboard-based input system**
- **Winner detection logic**
- **Draw condition checking**
- **Screen refresh and board update**
- **Replay functionality**

Technologies Used

The technologies and concepts used in this project are:

- **C++ Programming Language**
- **Console-based User Interface**
- **Arrays**
- **Functions**
- **Conditional Statements**
- **Loops**
- **Keyboard Input Handling**
- **Object-Oriented Programming Concepts**

Working of the Project

The project starts by initializing a 3×3 game board using a two-dimensional character array. Empty spaces on the board are represented using the '-' symbol.

The game operates in the following sequence:

1. **The game board is displayed on the console screen.**
2. **Players enter their moves using keyboard inputs.**
3. **Player X and Player O take turns alternately.**
4. **After every move, the board is updated and displayed again.**
5. **The program checks all rows, columns, and diagonals to determine the winner.**
6. **If all positions are filled without any winning condition, the game declares a draw.**
7. **The program also allows restarting the game after completion.**

The project uses separate functions for each operation, which improves program structure and code readability.

Main Functions Used in the Program

1. Draw()

This function is responsible for displaying the Tic Tac Toe board on the console screen. It continuously updates the board after every move and shows the current player's turn.

2. Input()

This function takes keyboard input from the user using _getch() and converts it into board positions. Specific numeric keypad values are mapped to board coordinates.

3. Set()

This function places the player’s symbol (X or O) on the selected position if the space is empty. It also switches the player turn after every valid move.

4. Logic()

This function checks all possible winning conditions including rows, columns, and diagonals. It also checks whether the match ends in a draw.

5. showResult()

This function displays the final result of the game such as:

- **Player X won**
- **Player O won**
- **Draw**

6. Game()

This function controls the overall game flow including board initialization, game execution, and replay functionality.

a. Implementation and Testing

The implementation of the project was completed using the C++ programming language in a console environment. Different programming concepts such as arrays, loops, functions, and conditions were integrated together to create the complete game structure.

The game board was implemented using a two-dimensional array, while multiple functions were used to handle input processing, board display, and result checking.

During the testing phase, the project was tested with different gameplay conditions to verify the correctness of the game logic. Multiple input combinations were used to ensure proper execution of winner detection and draw conditions.

The following testing activities were performed:

- **Testing player input handling**
- **Verifying turn switching**
- **Checking row winning conditions**
- **Checking column winning conditions**
- **Checking diagonal winning conditions**
- **Testing draw situations**
- **Verifying replay functionality**
- **Detecting and correcting logical errors**

The testing process ensured that the game functioned correctly under different scenarios.

b. Results and Discussions

The project was implemented successfully and achieved all intended objectives. The game operated correctly in the console environment and successfully handled player interaction, move validation, and winner detection.The project showed the practical application of C++ programming concepts and improved understanding of game logic. The use of separate functions improved code readability and program structure.The project also highlights the importance of logical thinking and debugging in software development. Continuous testing and improvements, the program was optimized for better execution and user interaction.

The final output of the project successfully displayed:

- **Player turns**
- **Updated game board**
- **Winning messages**
- **Draw condition**
- **Replay functionality**

The project proved to be an effective learning experience for understanding console application development and programming logic.

c. Conclusion and Future Scope

Conclusion

The Console Tic Tac Toe Using C++ project was completed successfully during the internship training period. The project helped in improving programming knowledge, logical thinking, and practical implementation skills.

Through this project, important concepts of C++ programming such as arrays, loops, functions, and conditional statements were understood and applied practically. The project also enhanced confidence in software development and debugging techniques.

Future Scope

The project can be improved further by adding advanced features and graphical enhancements in the future. Some possible future improvements are:

- **Single-player mode with computer AI**
- **Graphical User Interface (GUI)**
- **Score tracking system**
- **Difficulty level selection**
- **Database integration for saving scores**

These improvements can make the project more interactive and closer to real-world gaming applications.

d. Challenges and Solutions

During the development of the project, several challenges were faced while implementing the game logic and handling different gameplay conditions.

Challenges Faced

- **Understanding winner detection logic**
- **Managing player turns correctly**
- **Handling invalid inputs**
- **Debugging logical errors**
- **Implementing draw condition checking**
- **Updating the game board dynamically**

Solutions Applied

- **Game logic was divided into separate functions for easier management.**

- **Multiple testing scenarios were performed to identify errors.**
- **Conditional statements were optimized for proper result checking.**
- **Debugging techniques were used to remove syntax and logical issues.**

These solutions helped in the successful completion of the project.

e. Contribution in Project

The project was developed individually as a part of the internship training program. All programming, implementation, testing, debugging, and documentation work was completed independently under mentor supervision.

The major individual contributions in the project were:

- **Designing the game logic**
- **Writing the complete C++ code**
- **Implementing player input handling**
- **Creating the game board structure**
- **Implementing winner detection conditions**
- **Testing and debugging the program**
- **Improving code structure and execution flow**

The project significantly improved independent coding ability and practical understanding of software development concepts.

OUTPUT SCREENSHOTS:

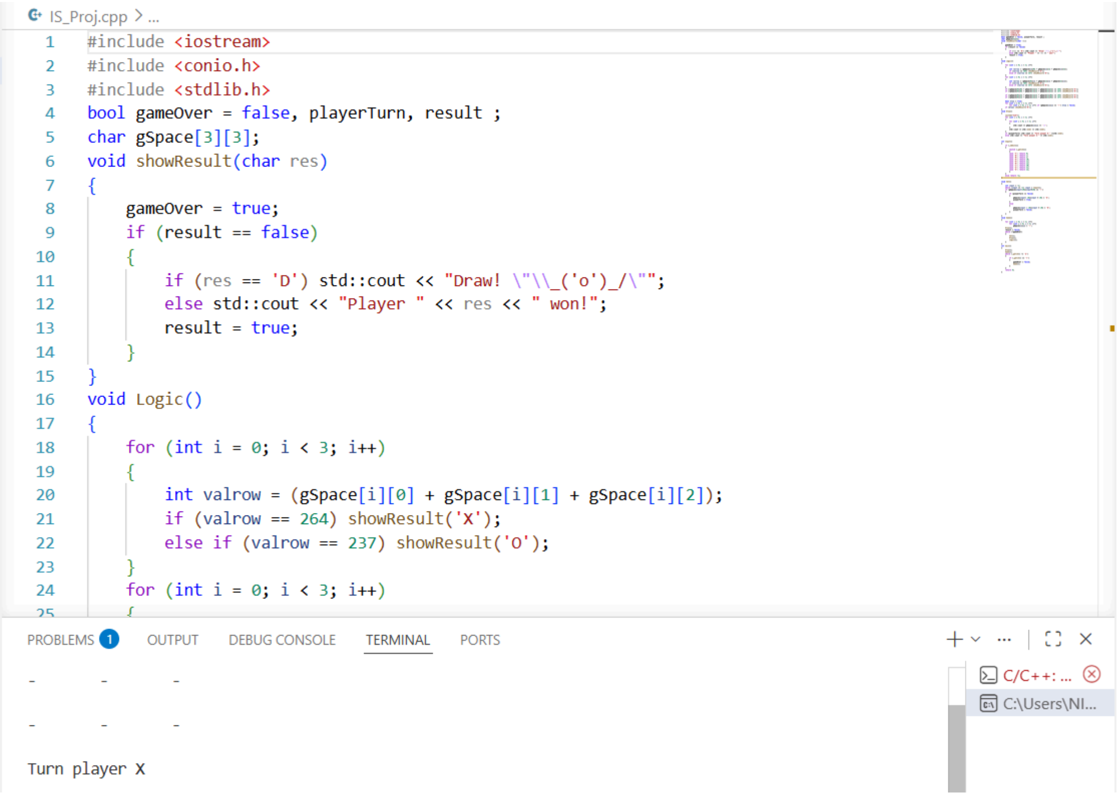

Fig:gameplay-1.png

Fig:gameplay-2.png

OUTPUT SCREENSHOTS:

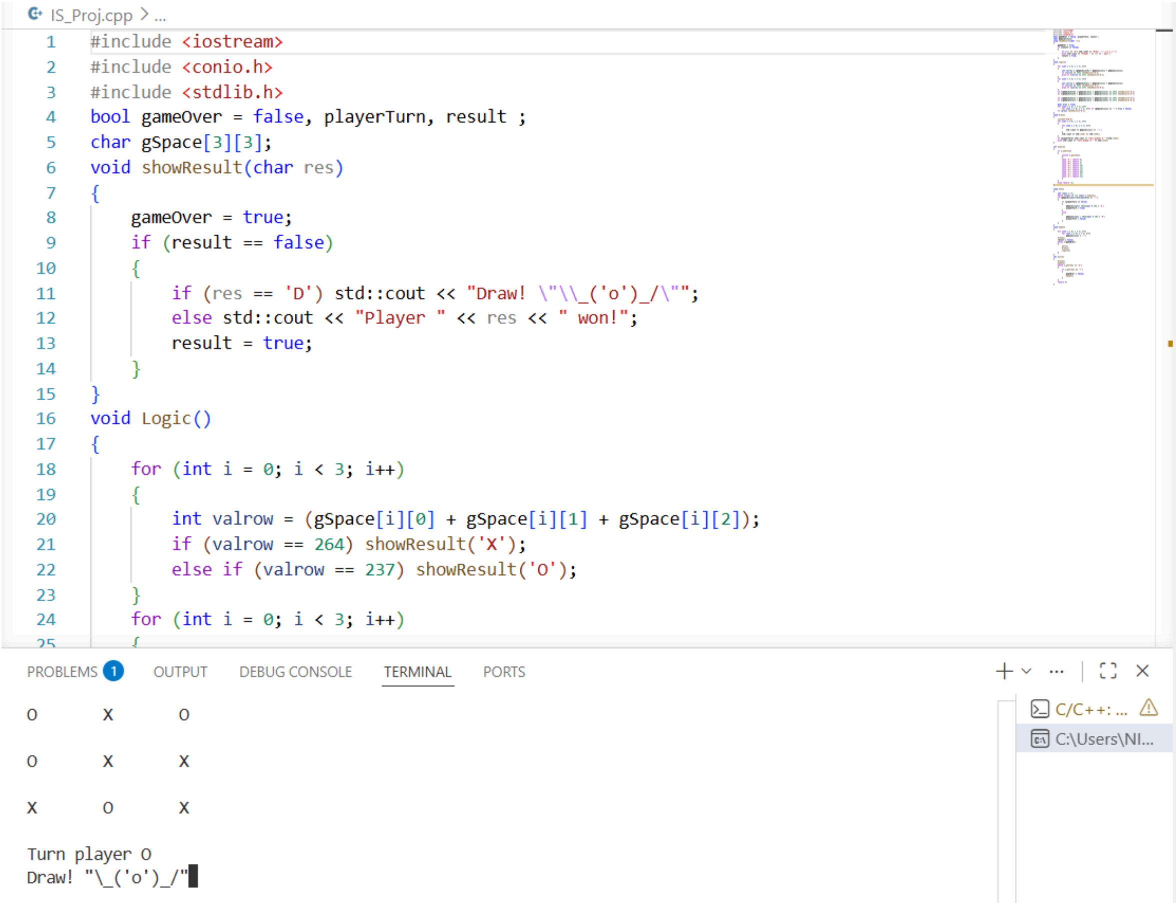

Fig:draw-screen.png

Fig:winner-screen.png

8. CONCLUSION

The internship at WingsIT Education and Solutions was a highly valuable learning experience that helped in improving technical knowledge, programming skills, and practical understanding of software development concepts. The internship provided an opportunity to apply theoretical concepts of C++ programming in real-world project implementation and enhanced overall confidence in coding and problem-solving.

During the internship period, various programming concepts such as loops, functions, arrays, conditional statements, and Object-Oriented Programming (OOP) principles were learned and implemented practically. Continuous coding practice and project development activities helped in strengthening logical thinking and debugging abilities.

The development of the project titled "Console Tic Tac Toe Using C++" played an important role in understanding the practical workflow of software development. The project successfully demonstrated the implementation of game logic, player interaction, winner detection, and console-based application design using C++ programming concepts.

The internship also improved self-learning capability, independent project development skills, and analytical thinking. Guidance from mentors and regular practical sessions helped in understanding programming concepts more effectively and solving technical challenges efficiently.

Overall, the internship was successful in achieving its objectives and provided meaningful exposure to real-time programming practices and software development methodologies. The knowledge and experience gained during this internship will be highly beneficial for future academic projects, advanced learning, and career development in the field of Computer Science and Software Engineering.

9. REFERENCES

The following references and resources were used during the internship training and development of the project “Console Tic Tac Toe Using C++”:

1. **Bjarne Stroustrup, *The C++ Programming Language*, Addison-Wesley Publication.**
2. **Herbert Schildt, *C++: The Complete Reference*, McGraw Hill Education.**
3. **E. Balagurusamy, *Object Oriented Programming with C++*, Tata McGraw Hill Publication.**
4. **Study materials and practical notes provided by WingsIT Education and Solutions.**
5. **Online C++ programming tutorials and coding references.**
6. **Documentation and examples related to:**
   - **Arrays**
   - **Functions**
   - **Loops**
   - **Conditional Statements**
   - **Object-Oriented Programming Concepts**
7. **Console programming practice programs and debugging exercises performed during internship training.**
8. **C++ Compiler and Development Environment used for coding, testing, and execution of the project.**
9. **Technical guidance and mentoring support provided by faculty members and trainers at WingsIT Education and Solutions.**
10. **Self-practice coding sessions and logical problem-solving exercises completed during the internship period.**

10. APPENDICES

Appendix – A

Internship Completion Certificate

The internship completion certificate issued by WingsIT Education and Solutions certifies the successful completion of internship training in the field of C++ Programming and Software Development during the specified training period.

The certificate confirms participation, project completion, and practical training received during the internship program.

Appendix – B

Project Source Code

The complete source code of the project “Console Tic Tac Toe Using C++” is attached in the appendix section. The source code includes:

- **Game board implementation**
- **Player input handling**
- **Winner detection logic**
- **Draw condition checking**
- **Game control functions and Replay functionality**

The source code demonstrates the use of C++ programming concepts such as arrays, loops, functions, conditional statements, and Object-Oriented Programming principles.

Appendix – C

Sample Output Screens

The output section contains sample execution screens of the Console Tic Tac Toe project including:

- **Initial game board display and Player turn interface**
- **Winning condition output**
- **Draw result display**

These outputs shows the successful execution of the program under different conditions.

Appendix – D

Tools and Software Used

The following tools and software were used during the internship and project development:

- **C++ Programming Language**
- **Console-based Development Environment**
- **C++ Compiler**
- **Code Editor / IDE**

Appendix – E

Learning Outcomes Summary

The internship and project development helped in improving the following areas:

- **C++ Programming Skills**
- **Object-Oriented Programming Concepts**
- **Logic Building and Problem Solving**
- **Debugging Techniques**
- **Console Application Development**
- **Independent Project Implementation**
- **Software Development Understanding**

Appendix – F

Project Flow Overview

The overall project workflow followed the following sequence:

1. **Understanding project requirements**
2. **Designing game logic**
3. **Writing C++ code**
4. **Implementing functions and conditions**
5. **Testing gameplay scenarios**
6. **Debugging and optimization**
7. **Final project execution and documentation**

The workflow ensured systematic project development and successful completion of the Console Tic Tac Toe project.